

MOTH — Race 7



Lago di Garda, Malcesine · 4 Jul 2026 · Start 16:41 (local)

Race course — all boats



Fleet

CRO 5087 *

AUS 5250 *

ESP 56

GER 5248

AUS 5120 *

Race winner

- CRO 5087 * finished first in this race.
- 2nd: AUS 5250 * 3rd: ESP 56
- Afternoon breeze; W-L-W-L course.
- Wind, marks and course axis estimated from GPS (no wind sensor in logs).

Instrument note (*)

- CRO 5087, AUS 5250, AUS 5120: speed sensor mounted rotated. Heel & heel-stability are rebuilt from the boat's gravity + course — an accurate estimate, not exact to the degree.

Positions at the marks — gains (green) / losses (red)

Finish	Boat	M1	M2	M3	Fin	Δ M1→M2	Δ M2→M3	Δ M3→Fin	Net
1	CRO 5087 *	2	2	1	1	+0	+1	+0	+1
2	AUS 5250 *	3	3	2	2	+0	+1	+0	+1
3	ESP 56	1	1	3	3	+0	-2	+0	-2
4	GER 5248	4	4	4	4	+0	+0	+0	+0
5	AUS 5120 *	5	5	5	5	+0	+0	+0	+0

M1 = windward 1 · M2 = leeward 1 · M3 = windward 2 · Fin = finish. Positive Δ (green) = places gained on that leg.



Fin	Boat	Line pos	Bias loss (m)	DTL gun	DTL -10s	SOG gun	SOG -10s	TWA gun	SOG 1'	Heel stab 1'	VMG 1'
1	CRO 5087 *	1-2	35	1	21	20.7	15.6	87	18.5	18.5	4.9
2	AUS 5250 *	2-2	23	9	34	21.6	19.3	66	19.5	4.4	11.2
3	ESP 56	1-2	34	1	32	18.7	19.3	78	16.9	8.4	8.8
4	GER 5248	1-1	41	62	93	15.3	15.9	68	16.8	14.2	6.7
5	AUS 5120 *	1-2	32	14	50	18.5	19.7	68	18.1	6.5	10.4

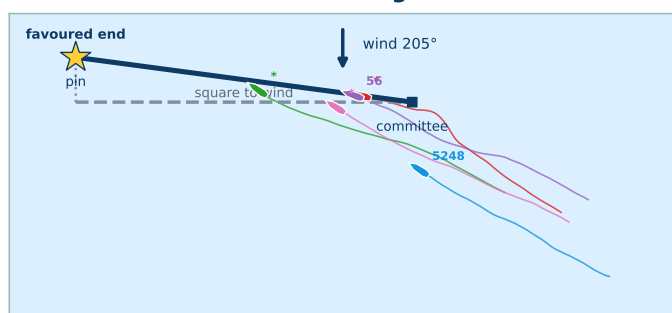
How to read the start

- Line pos: the line is split into 3 thirds, each third into 3 — read third-subpart from the committee end (1-1) to the pin end (3-3).
- Bias loss (m): metres given away at the gun from line position alone vs the favoured (pin) end, from the TWD bias.
- DTL = distance to line (m, negative = over early). 1' = first minute after the gun; heel stability = avg heel change per 10s (lower = steadier).

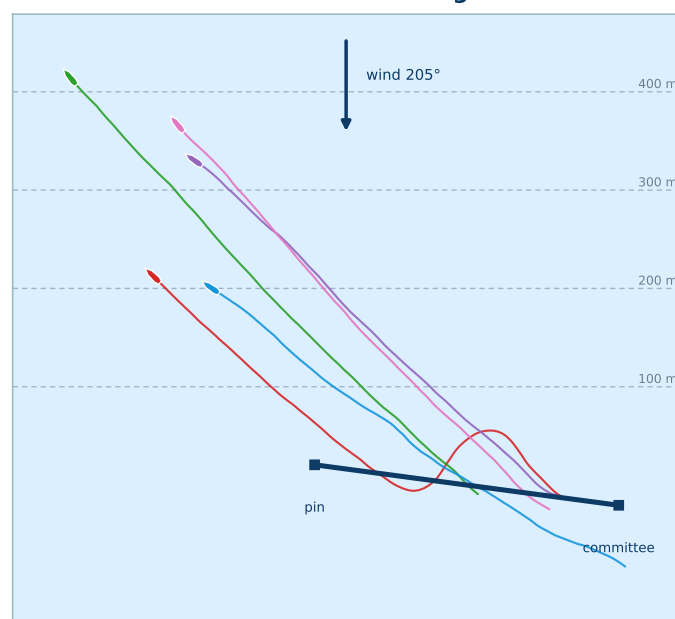
Line bias at the start (vs TWD)

- Start line 312 m, bearing 303°.
- Wind at start ≈ 205°, estimated from the first beat.
- Line bias vs TWD: 7.6° — Pin end favoured.
- Advantage ≈ 41 m (≈ 8.3 boat-lengths) for a boat starting at the pin end.

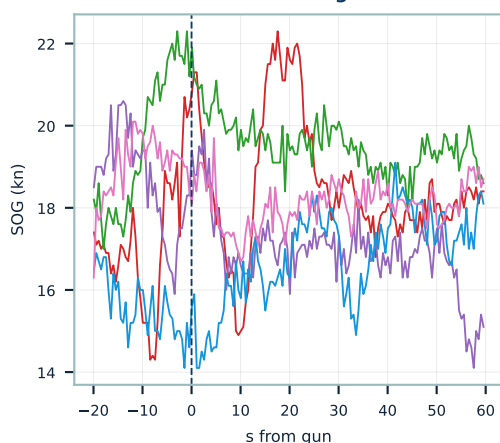
Position at gun



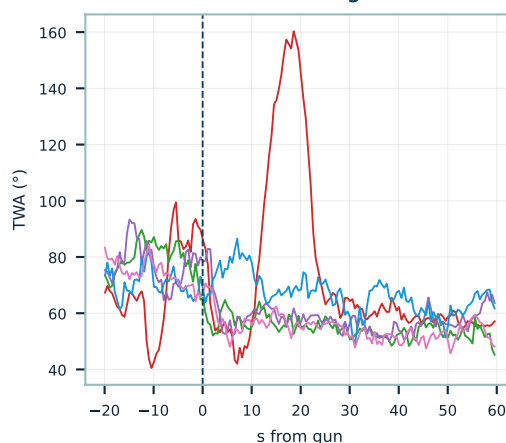
Position 1 min after gun



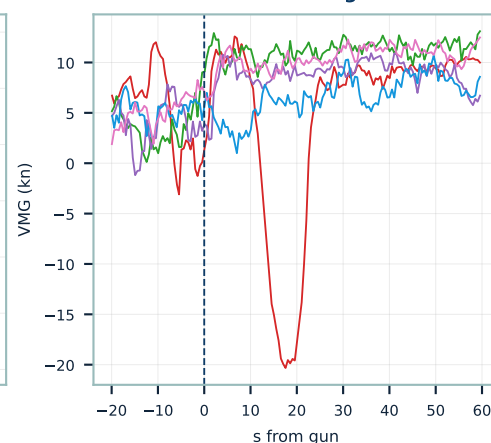
SOG around gun



TWA around gun



VMG around gun

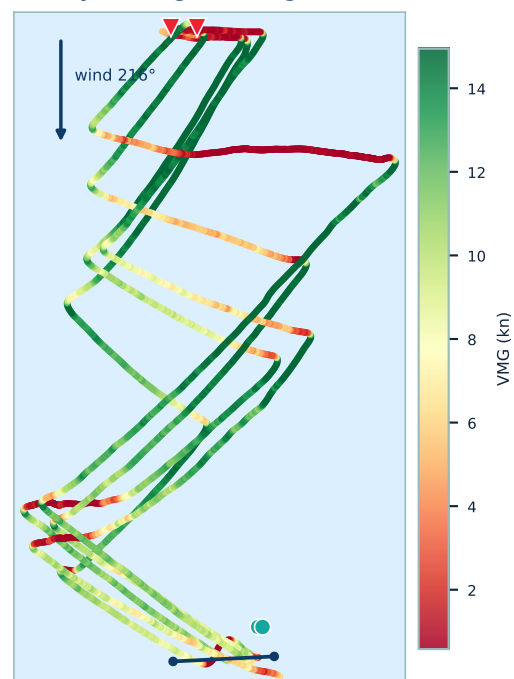




Tracks by boat



Tracks by VMG (green=high, red=low)



— CRO 5087* — AUS 5250* — ESP 56 — GER 5248 — AUS 5120*

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Tacks
3	ESP 56	10.15	53	17.26	19.80	22.6	10.6	+3.1	+1.1	3288	371	4
1	CRO 5087*	9.81	53	16.62	21.07	20.6	12.5	+4.2	+0.2	3258	382	4
2	AUS 5250*	9.08	58	16.27	20.90	20.4	14.6	-2.4	-0.1	3488	417	4
4	GER 5248	8.48	59	16.15	19.47	23.7	12.5	+2.5	+0.9	3792	457	4
5	AUS 5120*	7.31	65	14.00	21.90	17.6	10.1	+10.6	+0.2	3772	525	3

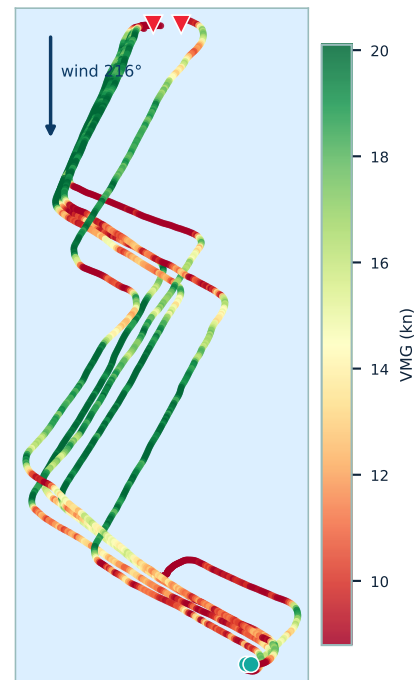
Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Tracks by boat



Tracks by VMG (green=high, red=low)



— CRO 5087* — AUS 5250* — ESP 56 — GER 5248 — AUS 5120*

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Gybes
2	AUS 5250*	15.49	138	21.97	24.30	14.4	6.8	-1.8	-0.2	2608	230	4
4	GER 5248	14.59	138	20.70	23.51	13.3	8.2	-1.4	+0.0	2633	247	4
5	AUS 5120*	14.57	137	20.66	23.50	10.5	8.4	+0.4	-0.3	2643	248	5
3	ESP 56	13.97	134	21.06	23.50	10.6	6.4	-1.8	-0.1	2778	256	4
1	CRO 5087*	13.68	130	20.75	25.26	14.1	7.9	-0.0	-0.3	2781	260	4

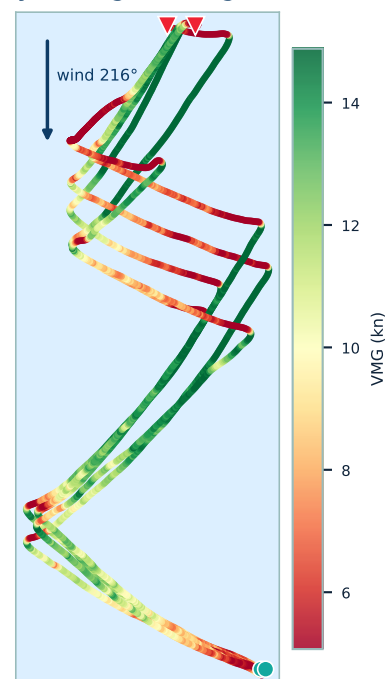
Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Tracks by boat



Tracks by VMG (green=high, red=low)



— CRO 5087* — AUS 5250* — ESP 56 — GER 5248 — AUS 5120*

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Tacks
1	CRO 5087*	11.58	47	17.60	20.40	21.0	7.6	+0.2	-0.0	2807	310	3
2	AUS 5250*	10.97	51	17.85	19.90	21.9	9.9	+1.6	-0.2	2998	327	4
4	GER 5248	10.72	48	16.58	18.70	24.6	8.5	+3.4	+0.4	2887	338	3
3	ESP 56	10.34	52	17.27	19.70	22.4	7.1	+1.6	+0.8	3083	348	3
5	AUS 5120*	8.86	56	16.09	20.80	21.9	14.7	-2.2	+0.5	3370	422	6

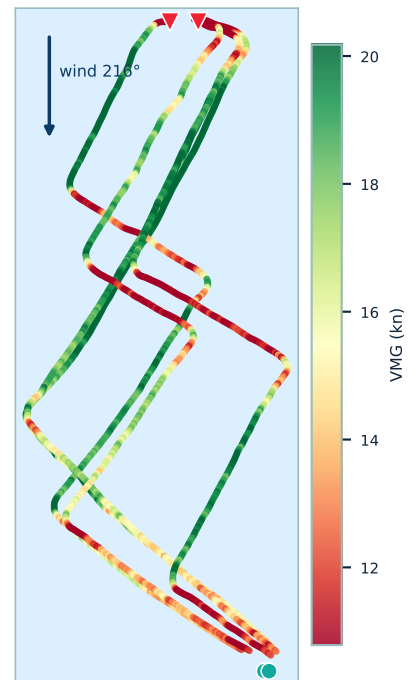
Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Tracks by boat



Tracks by VMG (green=high, red=low)



— CRO 5087* — AUS 5250* — ESP 56 — GER 5248 — AUS 5120*

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Gybes
1	CRO 5087*	17.69	141	23.44	26.32	14.3	10.1	+2.0	-1.3	2361	195	2
3	ESP 56	16.41	141	21.56	24.85	8.6	7.0	-1.4	-0.5	2348	211	2
2	AUS 5250*	16.37	137	23.14	26.45	15.2	6.8	-0.4	-0.5	2515	211	3
4	GER 5248	14.75	140	19.94	22.70	12.0	7.2	+2.1	-0.6	2440	237	4
5	AUS 5120*	14.46	138	20.23	24.04	10.2	7.0	+0.1	+0.1	2509	241	4

Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Finish	Boat	# Tacks	Entry	Exit	Min	SOG Δ	Angle Δ	Turn °/s	Heel in	Heel out	Dist loss (m)	VMG loss (m)
1	CRO 5087 *	7	17.7	13.4	11.1	6.6	+16	108	20	17	24.3	35.9
3	ESP 56	7	18.2	13.2	10.9	7.3	+12	110	21	18	27.8	32.3
4	GER 5248	7	17.5	12.6	10.9	6.7	+15	109	21	18	28.2	34.3
2	AUS 5250 *	8	18.5	13.1	11.2	7.3	+18	126	18	21	29.5	39.2
5	AUS 5120 *	9	16.8	10.8	8.9	7.9	+6	228	23	27	33.0	33.4

Rows ordered by Dist loss (most efficient first); the Finish column keeps each boat's race finish position.

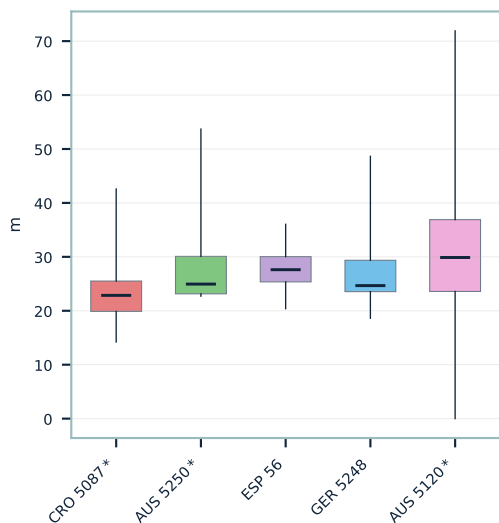
Tacks analysis — metric definitions

- Entry / Exit SOG = mean speed over 6 s before / 9 s after the manoeuvre. Min SOG = lowest speed in between.
- SOG Δ = Entry – Min (how much speed was bled off). Angle Δ = exit TWA – entry TWA (+ = exits wider).
- Rate of turn = peak heading change (°/s). Heel in/out = mean |heel|.
- VMG loss = made-good (toward the mark) deficit vs holding entry VMG. Dist loss = actual distance lost from the speed drop vs holding entry speed.

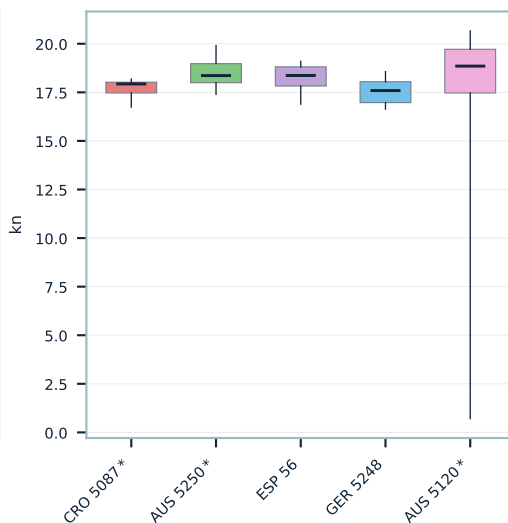
How to read the charts below

- Each box = one boat, ordered left→right by race finish. The plots span all of that boat's manoeuvres.
- Whiskers = min & max; box = middle 50% (25th–75th pct); line in the box = median. A lower, tighter box = more consistent.
- VMG/Dist loss, Speed loss, Min-speed: lower is better. Entry & Min speed: higher is better.

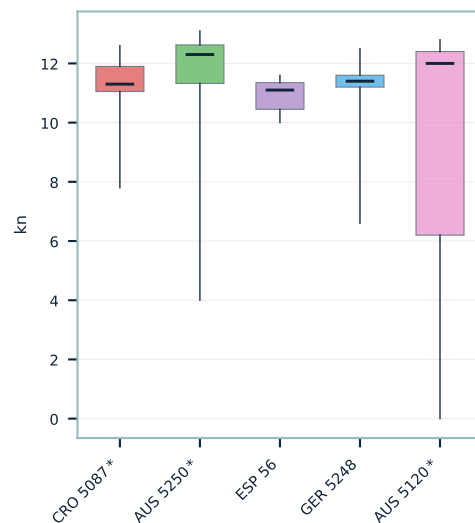
Dist loss (m, lower=better)



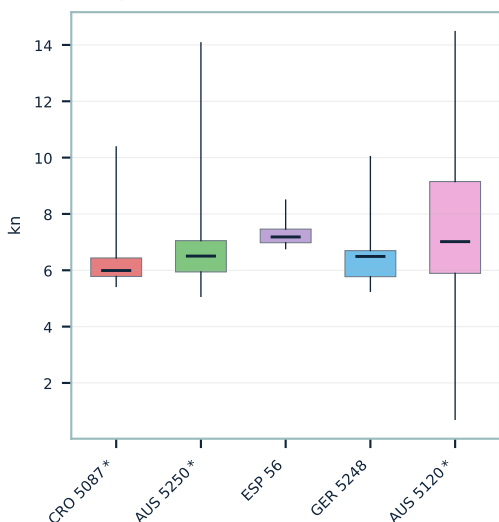
Entry speed (higher = better)



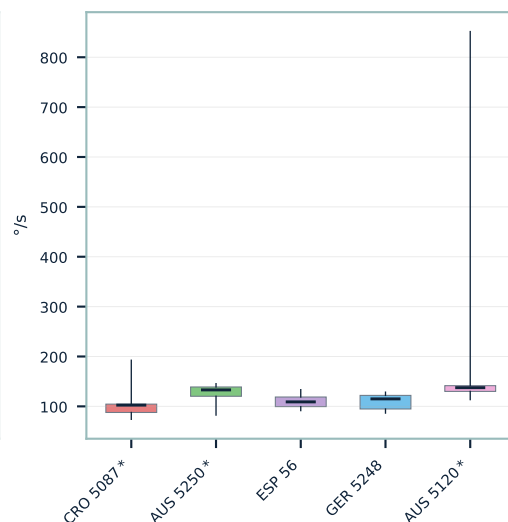
Min speed (higher = better)



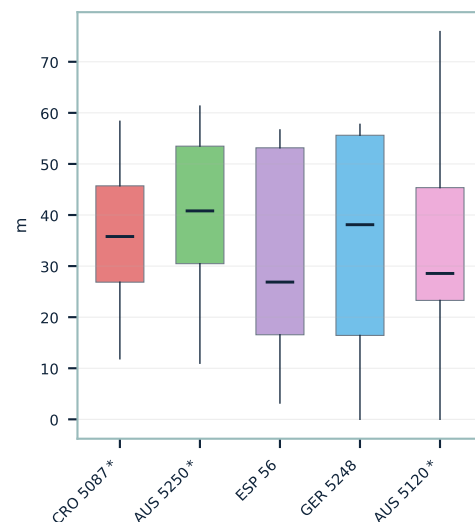
Speed loss SOG Δ (lower = better)



Rate of turn



VMG loss (m, lower=better)





Finish	Boat	# Gybes	Entry	Exit	Min	SOG Δ	Angle Δ	Turn °/s	Heel in	Heel out	Dist loss (m)	VMG loss (m)
1	CRO 5087 *	6	22.2	19.3	17.4	4.8	+1	69	14	9	16.2	12.9
2	AUS 5250 *	7	22.5	19.4	17.6	4.9	-6	76	14	11	17.0	20.8
3	ESP 56	6	20.7	17.5	15.6	5.1	-4	79	12	8	18.4	21.3
4	GER 5248	8	20.0	16.5	15.0	5.0	-8	75	11	9	19.5	23.9
5	AUS 5120 *	9	20.4	16.8	15.5	4.9	-1	74	10	8	20.3	16.5

Rows ordered by Dist loss (most efficient first); the Finish column keeps each boat's race finish position.

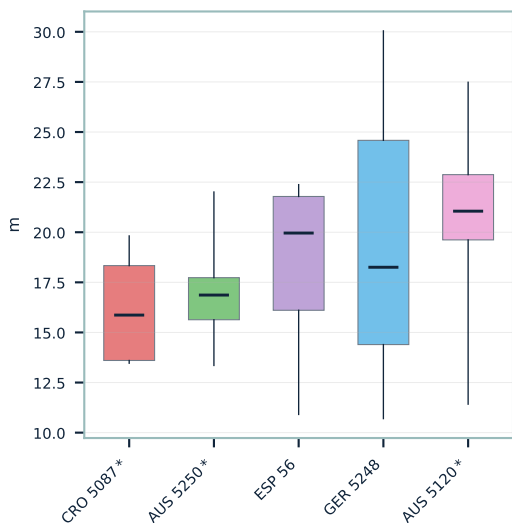
Gybes analysis — metric definitions

- Entry / Exit SOG = mean speed over 6 s before / 9 s after the manoeuvre. Min SOG = lowest speed in between.
- SOG Δ = Entry – Min (how much speed was bled off). Angle Δ = exit TWA – entry TWA (+ = exits wider).
- Rate of turn = peak heading change (°/s). Heel in/out = mean |heel|.
- VMG loss = made-good (toward the mark) deficit vs holding entry VMG. Dist loss = actual distance lost from the speed drop vs holding entry speed.

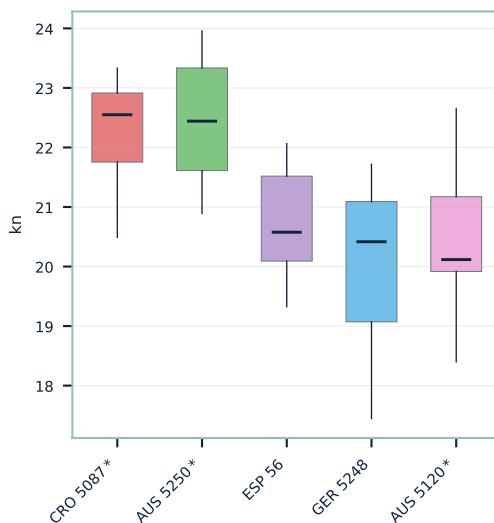
How to read the charts below

- Each box = one boat, ordered left→right by race finish. The plots span all of that boat's manoeuvres.
- Whiskers = min & max; box = middle 50% (25th–75th pct); line in the box = median. A lower, tighter box = more consistent.
- VMG/Dist loss, Speed loss, Min-speed: lower is better. Entry & Min speed: higher is better.

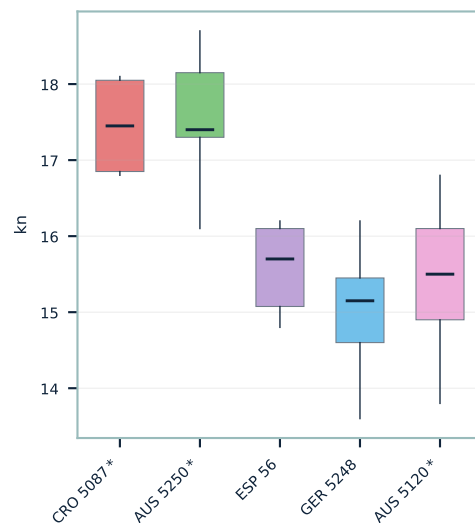
Dist loss (m, lower=better)



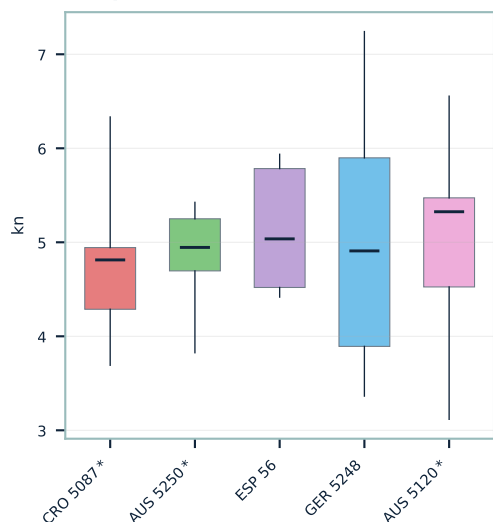
Entry speed (higher = better)



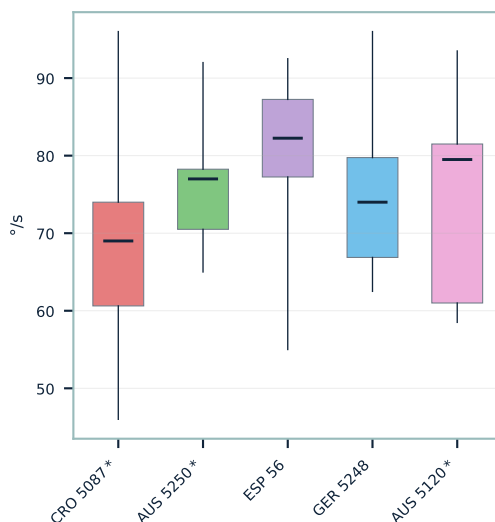
Min speed (higher = better)



Speed loss SOG Δ (lower = better)



Rate of turn



VMG loss (m, lower=better)

